

B.Tech/B.Tech (Hon.)/B. Tech-MS General Course Structure - Mathematics and Computing

Semester -I						Semester -II					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA 111	Discrete Mathematics	3	1	0	4	IMA 121/IMAXXX	Calculus and Linear Algebra/Linear Algebra and Applications	3	1	0	4
IEC 111	IT Workshop I	3	1	2	5	IMAXXX	Combinatorial Mathematics	3	0	0	3
ICS 112	Computer Programming	3	1	2	5	ICS 121	Data Structures I	3	1	2	5
IMAXXX	Real Analysis	3	1	0	4	ICS 122	Computer Organization	3	1	0	4
IHS 111	Communication Skills	3	0	0	3	IEC 122	IT Workshop II	2	1	2	4
IHS 112	Foreign Language	1	0	0	1	IHS 121	Personality Development	1	0	0	1
IPT 111	Physical Training I	0	0	2	0	IMAXXX	Graph Theory	3	1	0	4
Total		16	4	6	22	Total		18	5	4	25
Cumulative Credits at the End of First Year: 47											
Semester-III						Semester -IV					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA 211/IMAXXX	Probability, Statistics, and Random Processes/Probability and Statistics	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
IEC 211	Design and Analysis of Algorithms	3	1	0	4	IEC 221	Theory of Computation	3	1	0	4
ISC211	Introduction to Bioinformatics	2	0	0	2	IEC 222	Object-Oriented Analysis and Design	3	0	2	4
IMAXXX	Optimization: Theory and Applications	3	1	0	4	IMAXXX	Fundamentals of Data Science	1	0	2	2
IEC 214	IT Workshop III	2	1	2	4	ICS 224	Computer Networks	3	0	2	4
IMA XXX	Abstract Algebra and Cryptography	3	1	0	4	ICS 225	Data Structures III	1	0	2	2
ICS 215	Data Structures II	1	0	2	2	IHS 221	Fundamentals of Economics	1	0	0	1
IPT 211	Physical Training II	0	0	2	0	IHS 222	Principles of Management	1	0	0	1
Total		17	5	6	24	Total		16	2	8	22
Cumulative Credits at the End of Second Year: 93											
Semester -V						Semester -VI					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CSE311/MACXXX	Artificial Intelligence/Artificial Intelligence and Machine Learning	3	0	2	4	MACXXX	Mathematical and Computational Finance	3	1	0	4
MACXXX	Numerical Methods /Scientific Computing	3	0	2	4	ICS322/MACXXX	Large Language Models	3	0	2	4
IMA311	Soft Computing	3	0	0	3	MACXXX	Advanced Cryptography	3	0	2	4
IEC311/MACXXX	Digital Signal Processing/Digital Signal Processing and its Applications	3	0	2	4	MACXXX	Fluid Dynamics	3	1	0	4
IMA314	Optimization Techniques for Data Science	3	0	0	3	ISCXXX	Introduction to Network Science Analytics	3	0	0	3
						MACXXX	Complex Analysis	3	1	0	4
IMA313	Number Theory and Mathematical Theory of Coding	3	0	0	3						
Total		18	0	6	21	Total		18	3	4	23
Cumulative Credits at the End of Third Year: 137											
Semester-VII						Semester - VIII					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
MACXXX	Functional Analysis	3	1	0	4	MACXXX	Markov Decision Process and Reinforcement Learning	2	0	0	2
						ICS423	Internet of Things/Industrial Training	3	0	2	4
MACXXX	Topology/Algebraic Topology	3	1	0	4						
CSEXXX	Speech and Natural Language Processing/ Computer Vision: Principles and Process	3	0	2	4	MACXX	BTP II	6	0	0	6
MACXXX	BTP - I	6	0	0	6	ADS424	Honours Project Phase II (Optional)				
MAC XXX	Quantum Computing	3	1	0	4	MACXXX	Operation Research	3	0	0	3
Total		18	3	2	22	Total		14	0	2	15
Cumulative Credits at the End of Fourth Year: 174 (B. Tech)											
Semester -IX						Semester -X					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
ADS511	Research Project	12	0	0	12	ADS521	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (B. Tech-MS)											

BB.Tech/B.Tech (Hon.)/B. Tech-MS Detailed Course Structure - Mathematics and Computing

Semester -I					Semester -II						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA 111	Discrete Mathematics	3	1	0	4	IMA 121/IMAXXX	Calculus and Linear Algebra/Linear Algebra and Applications	3	1	0	4
IEC 111	IT Workshop I	3	1	2	5	IMAXXX	Combinatorial Mathematics	3	0	0	3
ICS 112	Computer Programming	3	1	2	5	ICS 121	Data Structures I	3	1	2	5
IMAXXX	Real Analysis	3	1	0	4	ICS 122	Computer Organization	3	1	0	4
IHS 111	Communication Skills	3	0	0	3	IEC 122	IT Workshop II	2	1	2	4
IHS 112	Foreign Language	1	0	0	1	IHS 121	Personality Development	1	0	0	1
IPT 111	Physical Training I	0	0	2	0	IMAXXX	Graph Theory	3	1	0	
Total		16	4	6	22	Total		18	5	4	25
Cumulative Credits at the End of First Year: 47											
Semester-III					Semester -IV						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA 211/IMAXXX	Probability, Statistics, and Random Processes/Probability and Statistics	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
IEC 211	Design and Analysis of Algorithms	3	1	0	4	IEC 221	Theory of Computation	3	1	0	4
ISC211	Introduction to Bioinformatics	2	0	0	2	IEC 222	Object-Oriented Analysis and Design	3	0	2	4
IMAXXX	Optimization: Theory and Applications	3	1	0	4	IMAXXX	Fundamentals of Data Science	1	0	2	2
IEC 214	IT Workshop III	2	1	2	4	ICS 224	Computer Networks	3	0	2	4
IMA XXX	Abstract Algebra and Cryptography	3	1	0	4	ICS 225	Data Structures III	1	0	2	2
ICS 215	Data Structures II	1	0	2	2	IHS 221	Fundamentals of Economics	1	0	0	1
IPT 211	Physical Training II	0	0	2	0	IHS 222	Principles of Management	1	0	0	1
Total		17	5	6	24	Total		16	2	8	22
Cumulative Credits at the End of Second Year: 93											
Semester -V					Semester -VI						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CSE311/MACXXX	Artificial Intelligence/Artificial Intelligence and Machine Learning	3	0	2	4	MACXXX	Mathematical and Computational Finance	3	1	0	4
MACXXX	Numerical Methods /Scientific Computing	3	0	2	4	ICS322/MACXXX	Large Language Models	3	0	2	4
IMA311	Soft Computing	3	0	0	3	MACXXX	Advanced Cryptography	3	0	2	4
IEC311/MACXXX	Digital Signal Processing/Digital Signal Processing and its Applications	3	0	2	4	MACXXX	Fluid Dynamics	3	1	0	4
IMA314	Optimization Techniques for Data Science	3	0	0	3	ISCXXX	Introduction to Network Science Analytics	3	0	0	3
						MACXXX	Complex Analysis	3	1	0	4
IMA313	Number Theory and Mathematical Theory of Coding	3	0	0	3						
Total		18	0	6	21	Total		18	3	4	23
Cumulative Credits at the End of Third Year: 137											
Semester-VII					Semester - VIII						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
MACXXX	Functional Analysis	3	1	0	4	MACXXX	Markov Decision Process and Reinforcement Learning	2	0	0	2
						ICS423	Internet of Things/Industrial Training	3	0	2	4
MACXXX	Topology/Algebraic Topology	3	1	0	4						
CSEXXX	Speech and Natural Language Processing/ Computer Vision: Principles and Process	3	0	2	4	MACXX	BTP II	6	0	0	6
MACXXX	BTP - I	6	0	0	6	ADS424	Honours Project Phase II (Optional)				
MAC XXX	Quantum Computing	3	1	0	4	MACXXX	Operation Research	3	0	0	3
Total		18	3	2	22	Total		14	0	2	15
Cumulative Credits at the End of Fourth Year: 174 (B. Tech)											
Semester -IX					Semester -X						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
ADS511	Research Project	12	0	0	12	ADS521	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (B. Tech-MS)											

Remark: To meet the minimum requirement of 186 credits for qualifying the B.Tech (Hon)Degree, students may take two additional projects of 6 credits each and, to meet the requirement of 218 credits for B.Tech-MS, students may take two additional projects of 6 credits each, two 4 credit research courses and 24 credit research project in addition to 174 credits requirement of B.Tech Degree.

Sr.	Code	Elective Courses	L	T	P	C
	MACXXX	Computational Algebra				
	MACXXX	Applied Linear Algebra and its Applications				
	MACXXX	Measure and Integration				
	MACXXX	Functional Analysis with Applications				
	MACXXX	Introduction to Differential Manifolds				
	MACXXX	Introduction to Stochastic Processes				
	MACXXX	Probability Theory				
	MACXXX	Intro. Dynamical Systems Theory				
	MACXXX	Mathematical Methods of Physics				
	MACXXX	Advanced Convex Optimization				
	MACXXX	Approximation Algorithm				
	MACXXX	Internet of Things				
	MACXXX	Fractional Calculus				
	MACXXX	Quantum Computing				
	MACXXX	Wavelet Theory/Algorithmic Graph Theory/Stochastic Processes and Applications.				
	MACXXX	Markov Decision Process and Reinforcement Learning				

B.TECH./B.TECH (HON)/DUAL DEGREE (B.TECH - MS) PROGRAMME

SI No	Course Description	Minimum Credits Requirement			Period
		BTech	BTech (Hon)	BTech-MS	
1	Institute Core courses	94	94	94	Semester I to IV
2	Bouquet Core Courses	24	24	24	Semester V to VIII
3	Stream Electives	18	18	18	Semester V to VIII
4	Institute Open Electives	9	9	9	Semester V to VIII
5	Humanities Electives	3	3	3	Semester V to VIII
6	Science Electives	3	3	3	Semester V to VIII
7	Maths Electives	3	3	3	Semester V to VIII
8	Engineering Elective	4	4	4	Semester V to VIII
9	Any other elective/Industrial Training	4	4	4	Semester V to VIII
10	BTech Projects	12	12	12	Semester VII to VIII
11	Honours Project		12	12	Semester VI to VIII
12	Research Courses			8	Semester VII to VIII
13	Research Project			24	Semester IX to X
Total Credits required for Successful Completion		174	186	218	
Minimum CGPA required for Successful Completion		5.5	8.0	8.0	